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Assignment -03
Image Processing and Pattern
Recognition
TCS-361

Q1.) Enlist the three types of discontinuity in digital image.

Ans.) In digital image processing, grey-level discontinuities are generally categorized into three types:

- Points: Isolated pixels that differ significantly in intensity from their surroundings.
- Lines: Discontinuities where intensity changes abruptly along a path with a certain width.
- Edges: The boundary between two regions with relatively distinct grey-level

properties.

Q2.) Describe texture and statistical approach with mathematical foundation.

Ans.) Texture refers to the spatial arrangement of intensity or color variations in an image or dataset. It captures properties like smoothness, coarseness, and regularity. A statistical approach models texture by analyzing pixel value distributions and their relationships.

First-order statistics consider individual pixel intensities (mean, variance). Second-order statistics examine relationships between pairs of pixels, often using gray-level co-occurrence matrices (GLCM).

For example, variance measures intensity dispersion:

$$\sigma^2 = (1/N) \sum_{i=1}^N \{I_i\}$$

$$(x_i - \mu)^2$$

Higher-order methods capture more complex dependencies. These statistical features form vectors used in classification, segmentation, or pattern recognition, providing a mathematical foundation for analyzing and distinguishing textures in data.

Q3.) Explain the role of Linear Algebra, Probability Theory, Expectation, Mean, Covariance, and Normal Distribution in Pattern Recognition.

Ans.) - Linear Algebra: Used for dimensionality reduction (e.g., PCA) and representing data as vectors/matrices for transformations.

- Probability Theory: Provides the framework for decision-making under uncertainty, such as Bayesian classification.

- Expectation (Mean): Acts as the

"prototype" or central representative of a class of patterns.

- Covariance: Describes how two variables change together, essential for understanding the "shape" of data clusters.

- Normal (Gaussian) Distribution: Often assumed for the distribution of features because it simplifies the math for many classifiers.

Q4.) A man is known to speak the truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.

Ans.) $P(A/B) = [P(B/A) P(A)] / [P(B/A) P(A) + P(B/\text{not } A) P(\text{not } A)]$

$$= [(3/4)(1/6)] / [(3/4)(1/6) + (1/4)(5/6)]$$

$$= (3/24) / (3/24 + 5/24)$$

$$= 3/8$$